ModelIt!

Student Activity Pack

Why Do Marbles Crash and Bounce?

The Surprising Science of Energy in Collisions

Grade 4 | NGSS 4-PS3-3 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens when you roll a marble into another marble that is sitting still?

2 Does a fast marble or a slow marble make a bigger crash? Why?

3 Where does the energy go when two things bump into each other?

4 Draw what you think happens when a fast marble hits a still marble.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Marble Speed | Collision Force | Energy Transfer

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Gentle Tap

Low marble speed

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Fast Roll

High marble speed

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Speed Comparison

Compare low and high speed collisions

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Marble Size — Bigger, heavier marbles have more momentum and transfer more energy in collisions

Surface Friction — Rough surfaces slow marbles down and reduce the energy available for collision

Angle of Impact — Head-on collisions transfer more energy than glancing collisions that just barely touch

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When two marbles crash into each other, where does the energy go? Use evidence from your simulation to support your answers.

1 Why does the first marble slow down after hitting the second marble?

2 If energy cannot be created or destroyed, where does the click sound come from during a marble collision?

3 How are marble collisions similar to what happens when a baseball bat hits a ball?

4 What would happen if you rolled two marbles at each other at the same speed?

5 How does your model help explain what happens in car crashes?

8 STEM Challenge: Design the Ultimate Marble Run

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YOUR ENGINEERING MISSION

Design a marble run course that uses collisions to move marbles through a series of obstacles and hit a target at the end.

THE SCENARIO

A toy company is holding a design competition for the best marble run. Your engineering team needs to design a course where one marble starts at the top and uses collisions to transfer energy through multiple marbles to hit a target cup at the end. The most accurate and creative design wins!

Engineering Design Thinking

1. WHAT MATERIALS WILL YOU USE TO BUILD YOUR MARBLE RUN TRACK?
2. HOW HIGH WILL YOUR STARTING RAMP BE TO GIVE THE FIRST MARBLE ENOUGH ENERGY?
3. HOW WILL YOU AIM THE COLLISIONS SO ENERGY TRANSFERS IN THE RIGHT DIRECTION?
4. HOW WILL YOU TEST AND IMPROVE YOUR DESIGN AFTER EACH ATTEMPT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Mechanical Engineers and Crash Safety Scientists use the science of collisions to design safer cars, better sports equipment, and more efficient machines. They earn $65,000–$120,000/year.